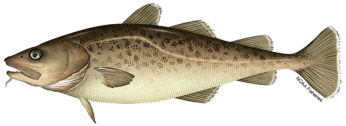
Pacific cod ‘Omics projects: Juvenile temperature responses, and more!



NOAA AFSC Behavioral Ecology Program at Hatfield
March 2026, Newport OR
Laura Spencer, Univ. of Washington SAFS & NOAA AFSC

Ben Laurel, NOAA AFSC Newport, OR
Louise Copeman, NOAA AFSC Newport, OR
Tom Hurst, NOAA AFSC Newport, OR
Mary Beth Rew-Hicks, NOAA AFSC Newport, OR
Paul Iseri, NOAA AFSC Newport, OR
Ingrid Spies, NOAA AFSC Seattle, WA
Steven Roberts, University of Washington SAFS



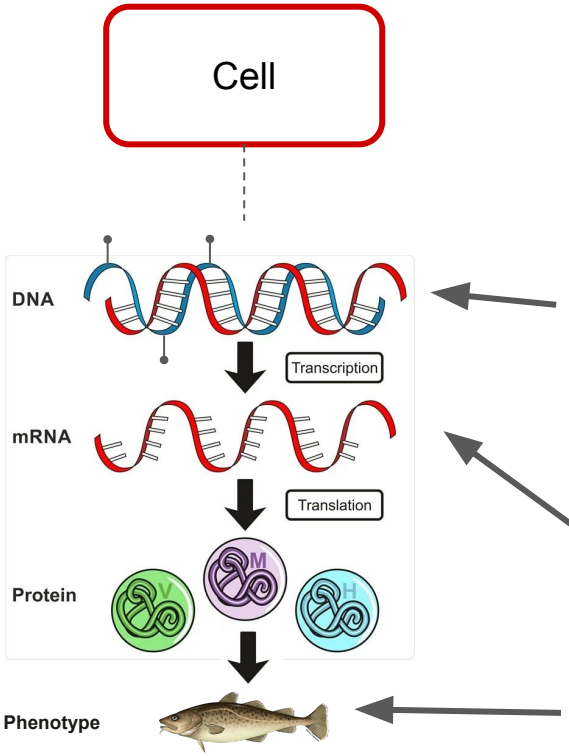
Today:

1. Juvenile Pacific cod temperature experiment

2. Molecular data - what is it, what does it looks like, and how are we using it?

3. Other Pacific cod genetics work, ongoing and future

What is molecular data?



Adapted from: udaix/Shutterstock.com

Appr. genome sizes (base pairs)

Pacific cod: 556 million
Humans: 3.2 billion
Snow: 2.0 billion
Tanner: 1.7 billion
Red king crab: 3.8 billion
Dungeness: 660 million

E.g. DNA sequence data - Who are they?

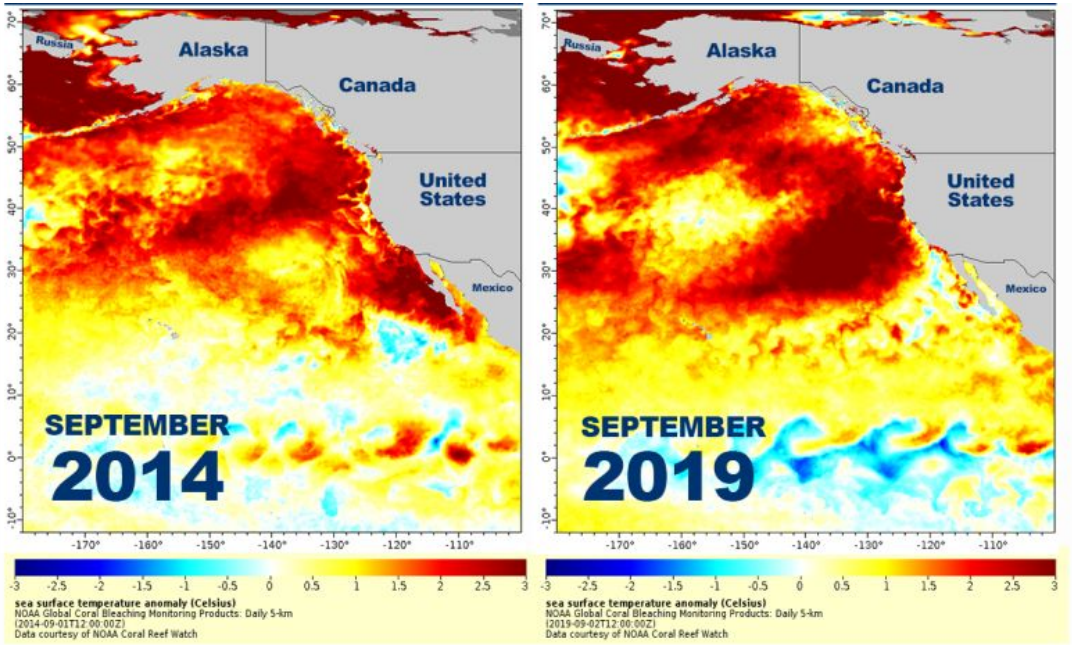
TGCCGGGTCCAGTTCGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT
TGCTGGGTCCAGTTCGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT

E.g. RNA sequence data - What are they doing right now?

UGCCGGGUCCAGUUCGCUGGGCUUUCGGCGCCUCGAGCUGACGGAAAGAU
UGCUGGGUCCAGUUCGCUGGGCUUUCGGCGCCUCGAGCUGACGGAAAGAT

E.g. source population, growth, survival, etc.

Heatwaves!



[NOAA feature story](), *September 05, 2019*



Pacific cod, *Gadus macrocephalus*

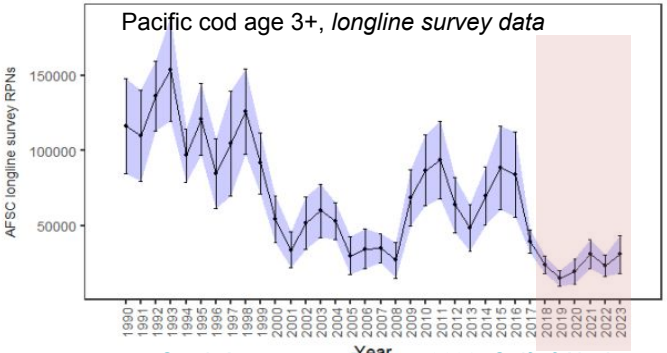
Extremely low cod numbers lead feds to close the Gulf of Alaska fishery for the first time

Alaska Public Media
Published December 6, 2019 at 8:22 PM AKST



Pacific cod (Photo Courtesy of Holland Dotts & the Alaska Marine Conservation Council)

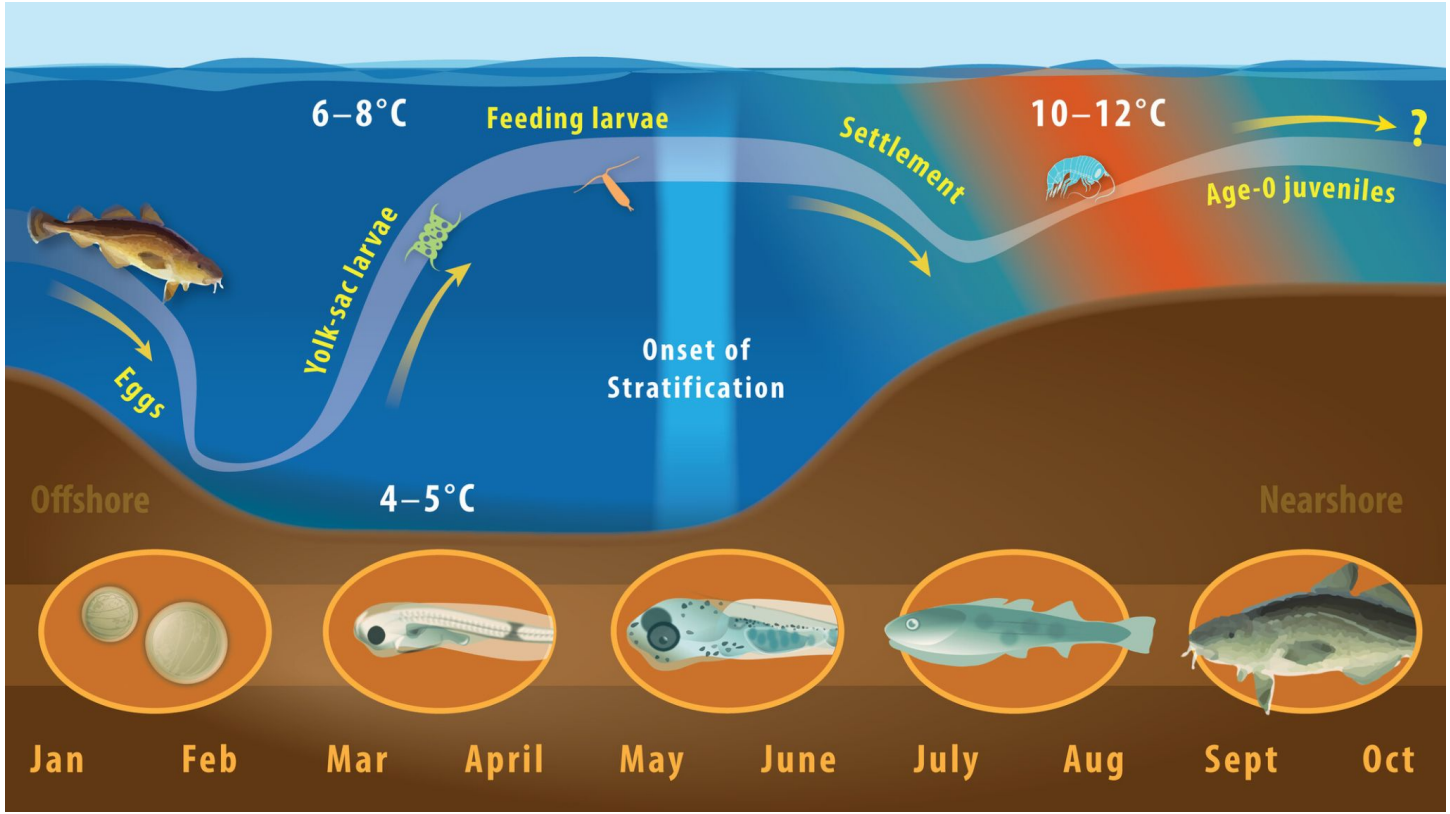
[Alaska Public Media](), *December 06, 2019*



[Stock Assessment Report, 2023, Gulf of Alaska](),



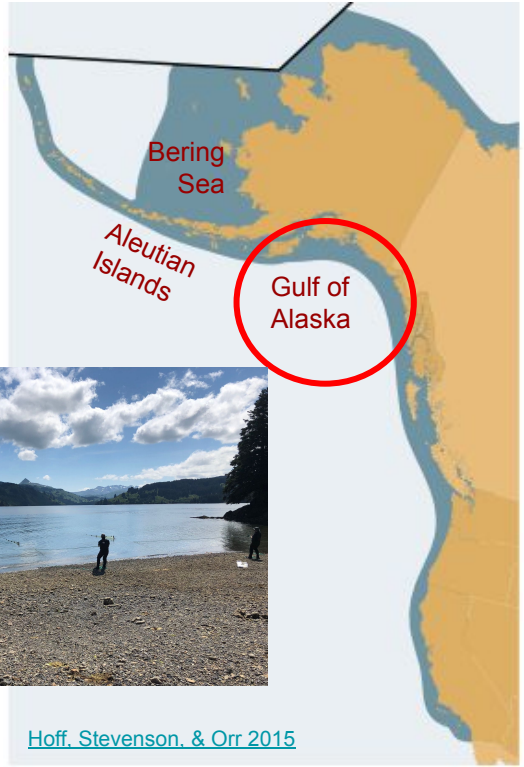
Pacific cod early life stages are very sensitive to temperature



Laurel et al. 2023, *in Fish & Fisheries*



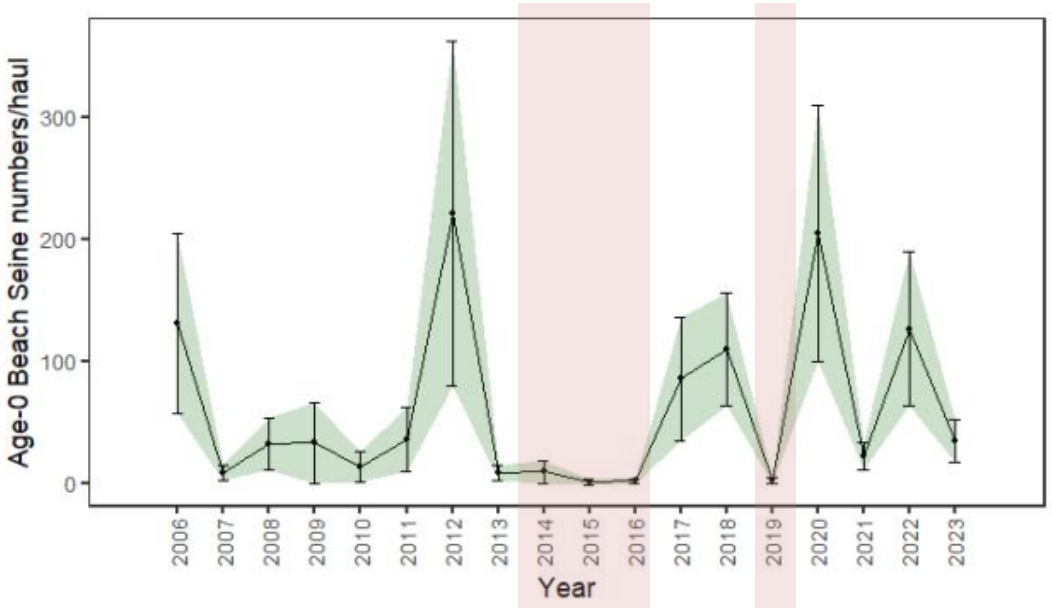
Low abundances likely related to low recruitment during heat waves





Hoff, Stevenson, & Orr 2015

Pacific cod juvenile (age-0) *beach seine survey CPUE*

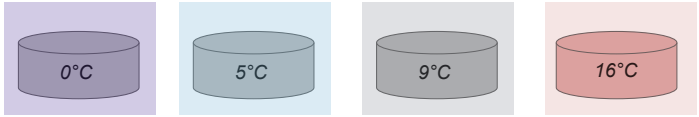


Stock Assessment Report, 2023, Gulf of Alaska,

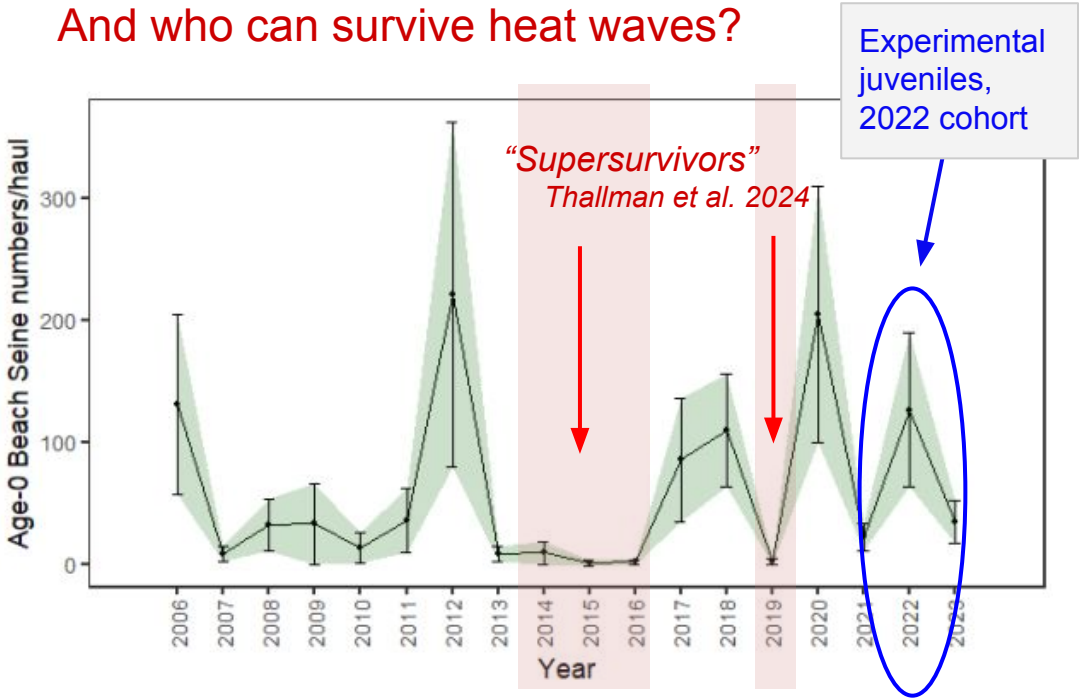


Question: How does temperature affect juvenile Pacific cod?





And who can survive heat waves?



Stock Assessment Report, 2023, Gulf of Alaska,



“Genome to Phenome” dataset across temperatures for Pacific cod





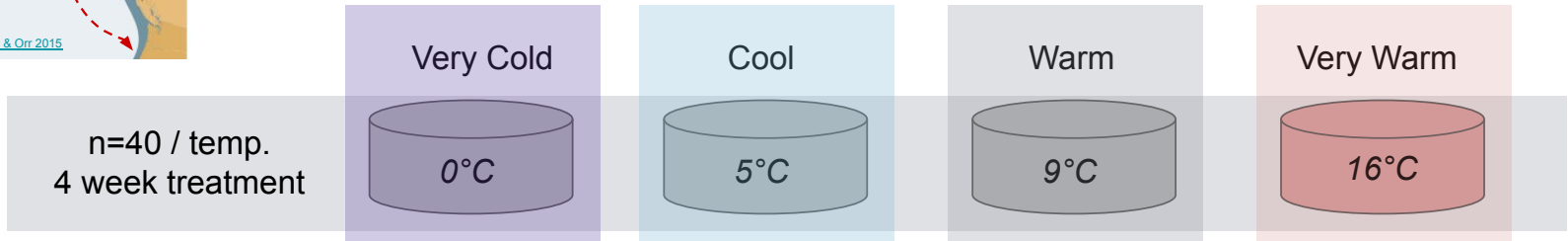
→ Individuals were tagged ←

a. Genetics (lcWGS)

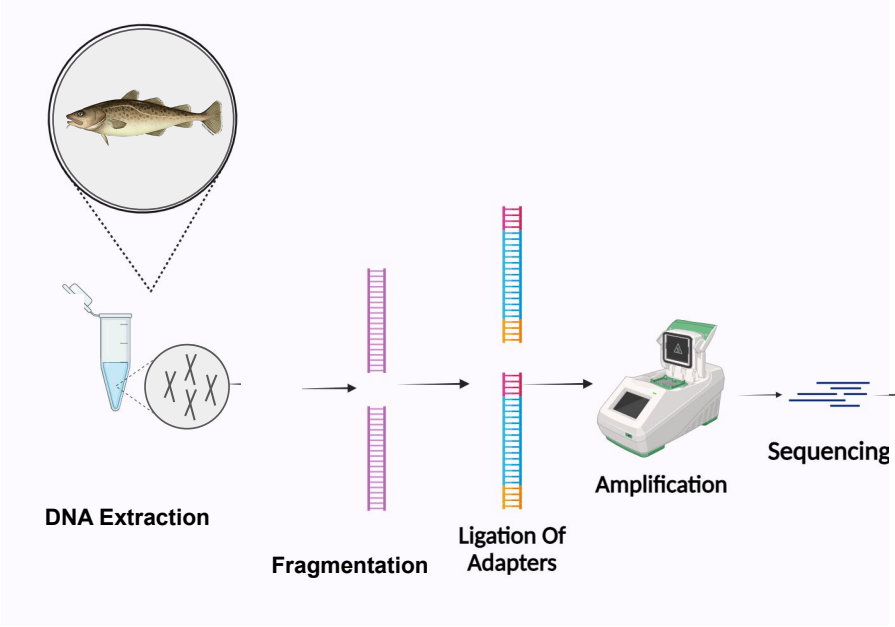
b. Gene expression in liver, n=18/temp (RNASeq)

c. Fitness Traits
   - Growth rates
   - Liver & Body Condition
   - Liver lipid concentrations (n=25/temp)
   - Survival





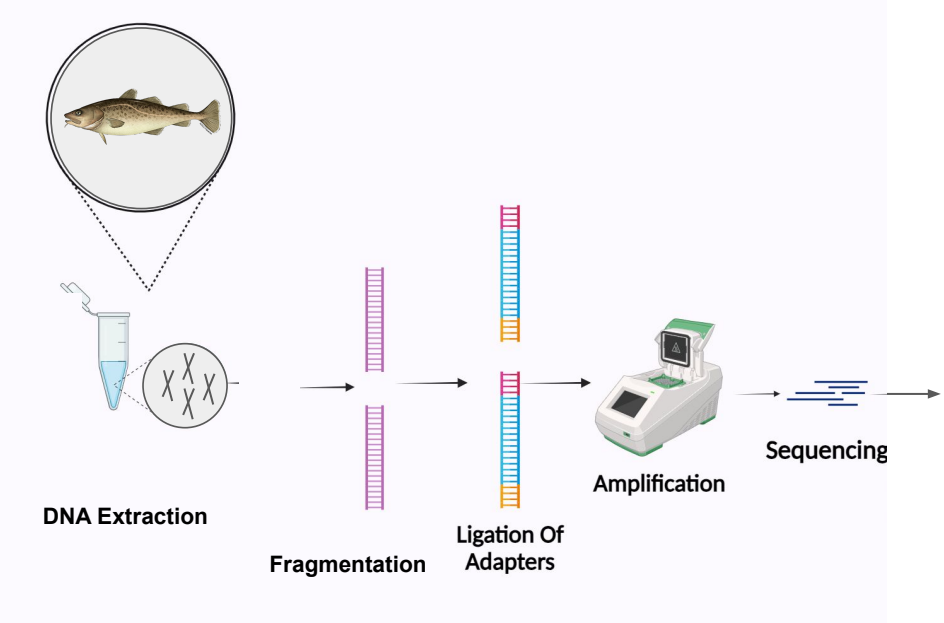
How do we get genetic data, and what does it look like?





Adapted from an image by Shraddha Tomar, BioRender

How do we get genetic data, and what does it look like?



Adapted from an image by Shraddha Tomar, BioRender

Reference sequence (e.g. genome)

TGCCGGGTCCAGTTCGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT

Individual A

TGCCGGGTCCAGTACGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT

Individual B

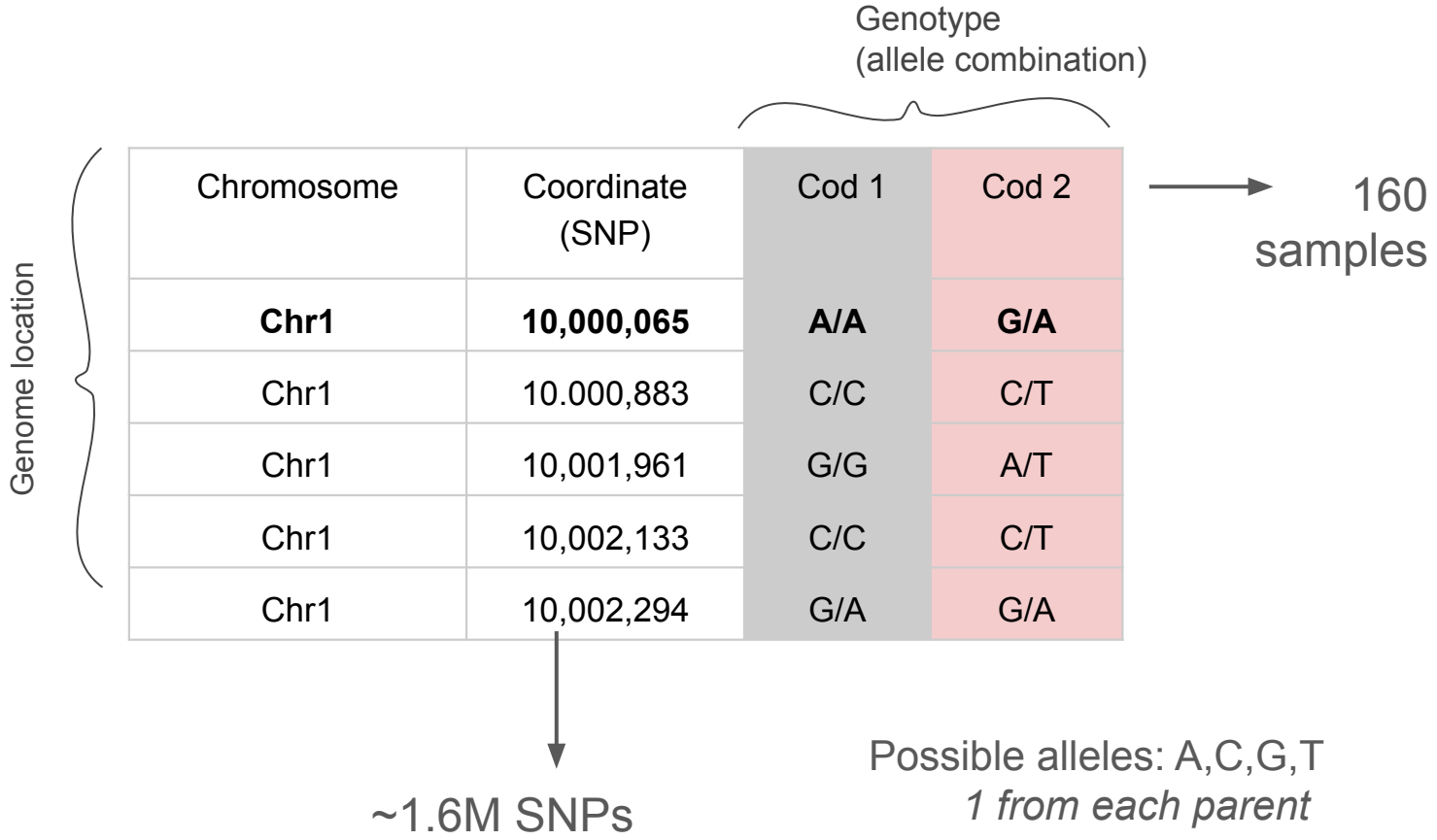
TGCCGGGTCCAGTACGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT

Individual C

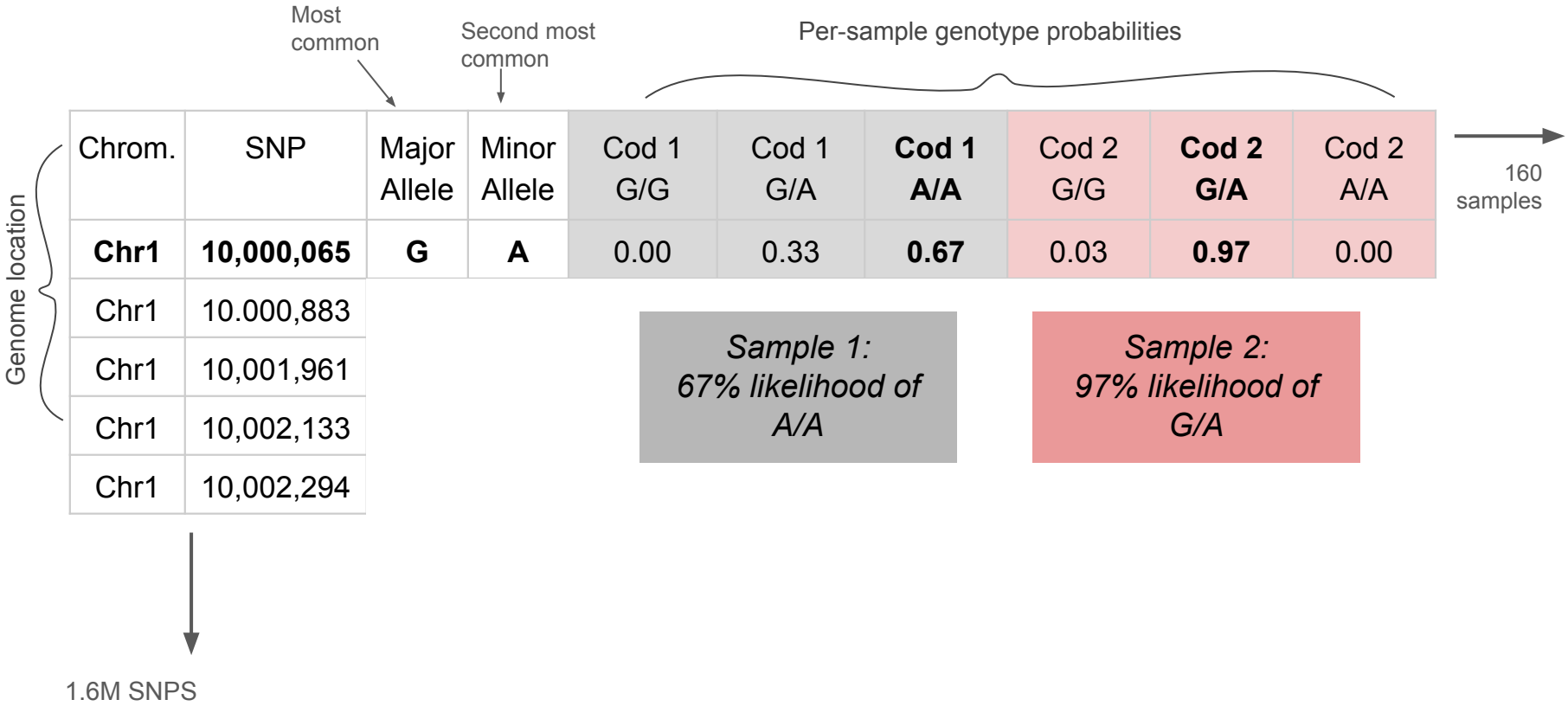
TGCCGGGTCCAGTTCGCTGGGCTTTCGGCGCCTCGAGCTGACGGAAAGAT

Single nucleotide polymorphisms (SNPs)

What does genetic data look like?

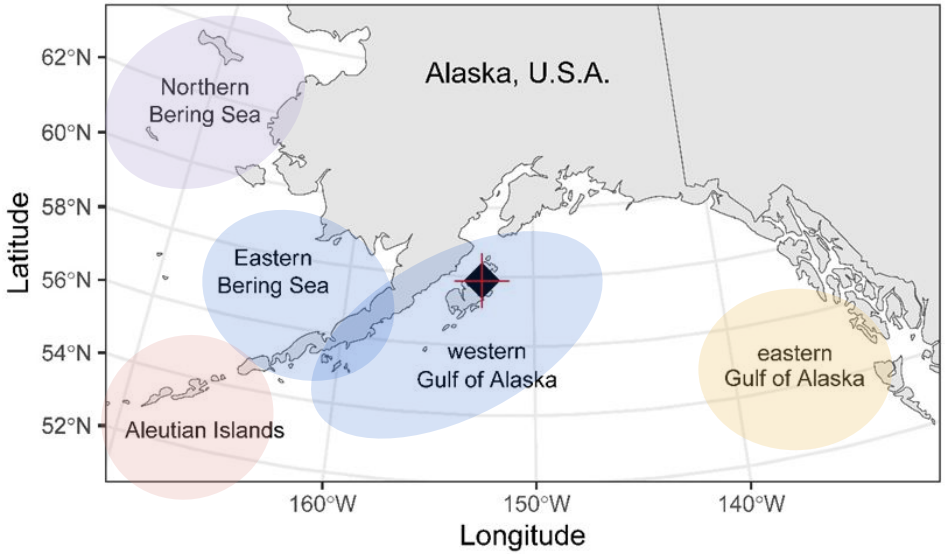


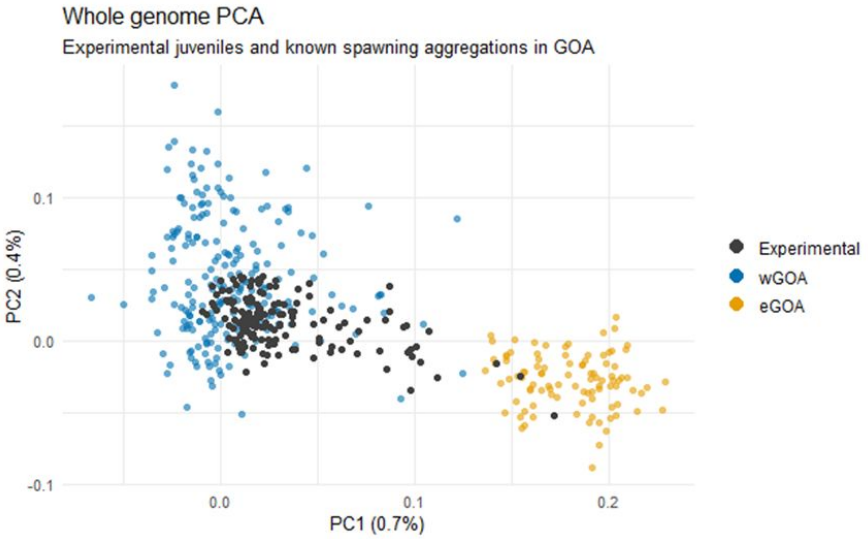
What does genotype likelihood data look like?



Genetics was used to identify population represented in experiment.

Most juveniles from west/central Gulf of Alaska, some with eastern GOA ancestry





Schaal et al. 2026, https://doi.org/10.1111/eva.70174



“Genome to Phenome” dataset across temperatures for Pacific cod





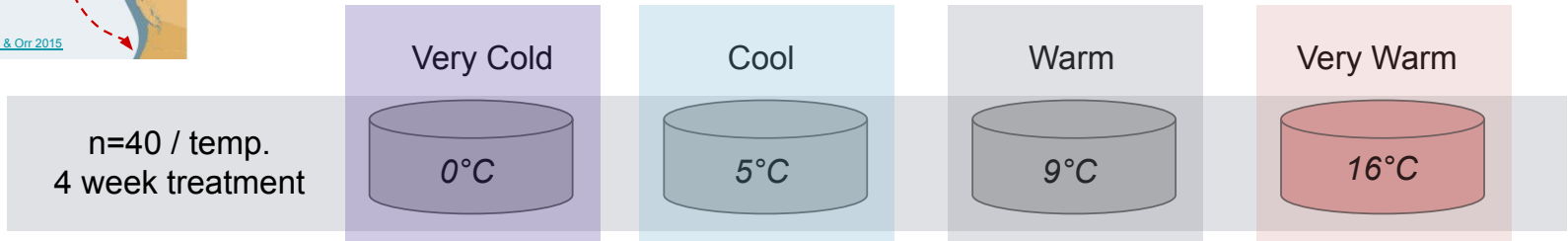
→ Individuals were tagged ←

a. Genetics (lcWGS)

b. Gene expression in liver, n=18/temp (RNASeq)

c. Fitness Traits
   - Growth rates
   - Liver & Body Condition
   - Liver lipid concentrations (n=25/temp)
   - Survival

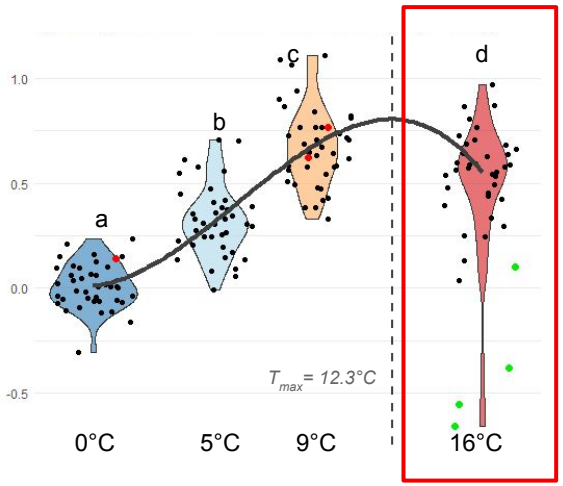




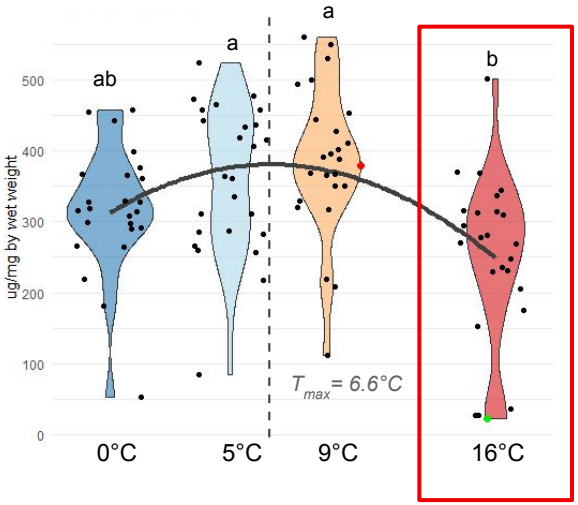


Warming slows growth, reduces energy reserves (lipid), & increases mortality - likely to impact overwinter survival.

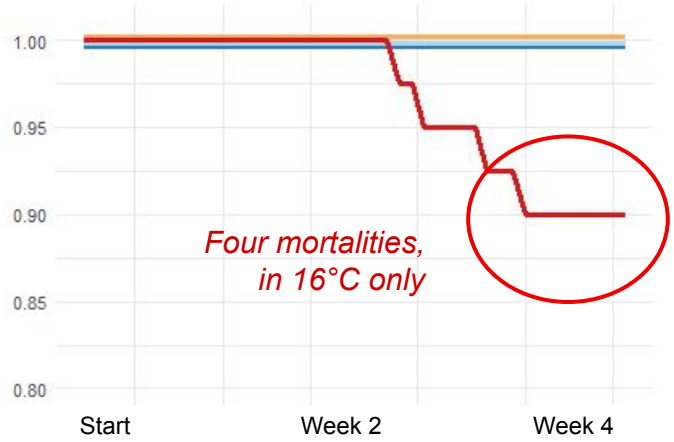
Growth Rate (weight, %/day)



Lipid Content in liver (mg)



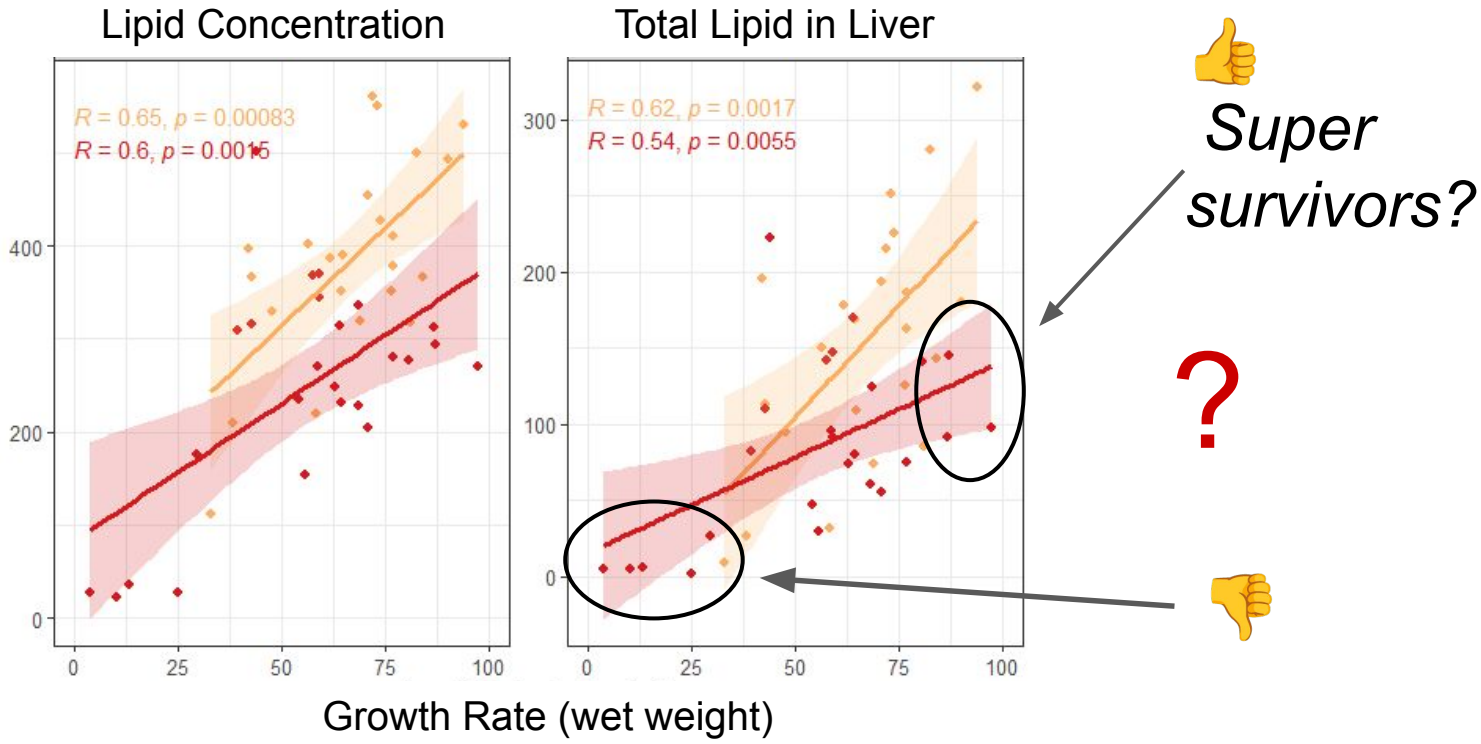
Survival



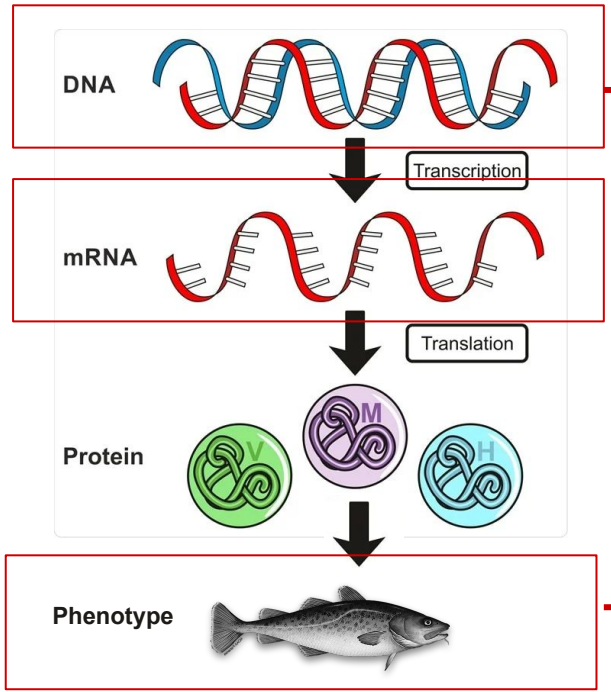
n=40/temperature



The fastest growers in warm temperatures also have the most energy reserves (lipid) in liver



“Genome to Phenome” dataset across temperatures



Adapted from: udaix/Shutterstock.com

→ Individuals were tagged! ←

a. Genetics (lcWGS)
b. Gene expression in liver (RNASeq)
c. Fitness Traits
   - Growth rate
   - Liver & Body Condition
   - Liver lipid concentration, content
   - Survival

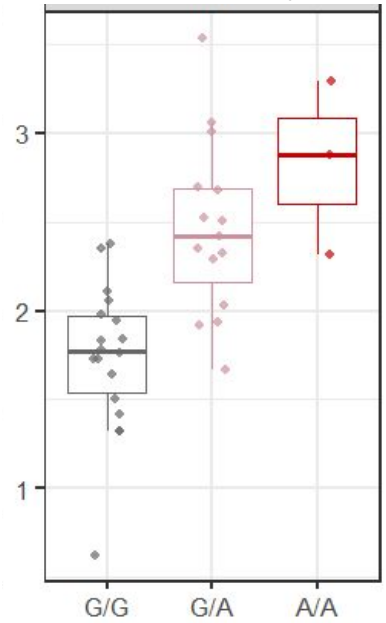
Which **sites on genome** are associated with **fitness traits?**

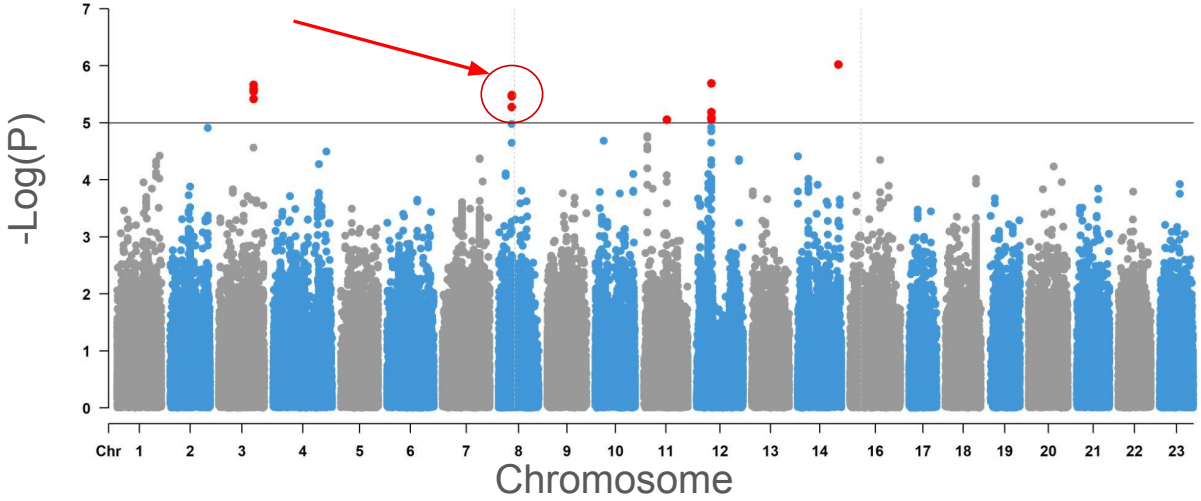
Genome-Wide Association Studies (GWAS) using single nucleotide polymorphisms (SNPs)



12 markers in 6 genes putatively associated with **liver condition (HSI)** in Pacific cod juveniles held in 16°C

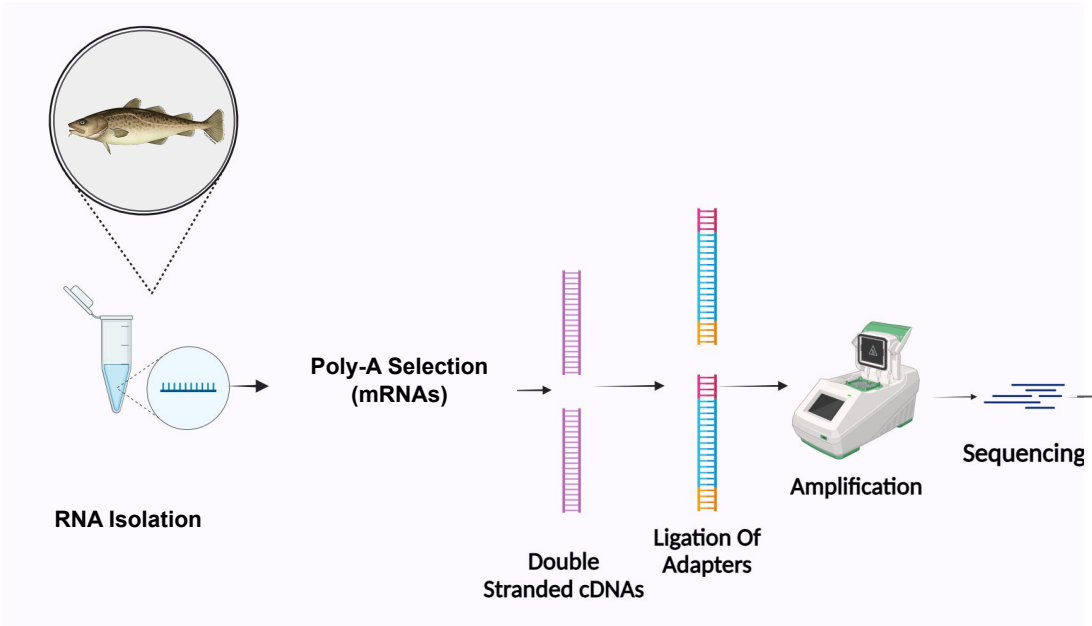
HSI by genotype (most likely)

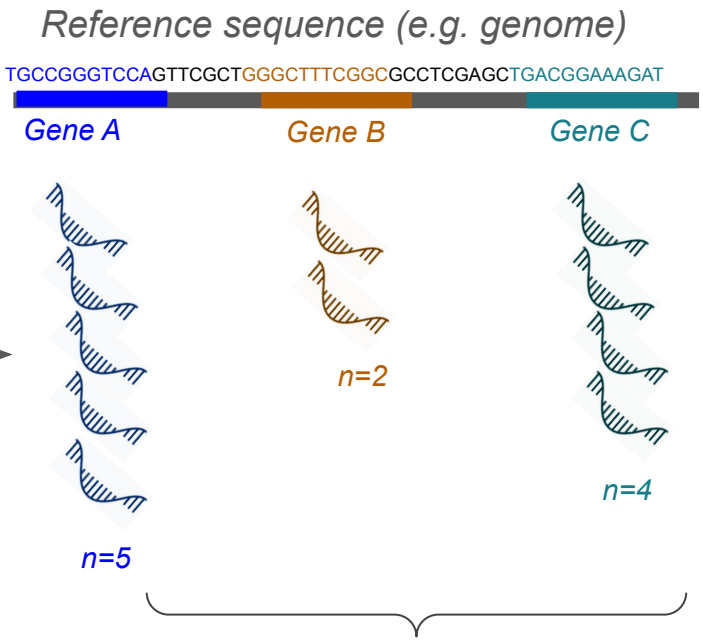




We also have expression data!

How do we get gene expression data, and what does it look like?





Adapted from an image by Shraddha Tomar, BioRender

Gene count matrix

How do we get gene expression data, and what does it look like?

Samples

Treatment *Control*

Genes

gene_gadmor	PCG001	PCG004	PCG011	PCG015	PCG017	PCG020	PCG029	PCG035
ND1	8276	8202	8327	7750	5508	4351	5673	6385
ND2	39142	43297	57032	31681	33489	30275	38921	36460
COX1	555463	631876	917827	551062	378628	403956	438595	455514
COX2	315625	309958	493176	294979	189346	229132	227614	219894
ATP6	82892	105415	89100	73253	77770	58061	86703	78859
COX3	165275	189321	235193	135559	112870	81377	135462	126615
ND3	10710	13595	10404	7013	10244	2567	9792	8231
ND4L	19364	31196	42855	16261	20566	17877	21021	16269
ND4	37081	42648	67118	30275	29274	37939	35681	34155
ND5	16299	21803	2823[illegible]			19960	19262	18187
ND6	4102	3787	318[illegible]			1774	3772	3391
CYTB	176843	211635	277395			114784	137070	144369
LOC115539476	260	228	333	219	270	266	312	266
LOC115539709	890	849	919	777	561	461	945	1000
LOC115538781	586	596	576	630	417	450	704	730
abhd14a	1197	1381	1629	1031	774	757	1158	1108
acy1	1626	1244	1670	1245	1224	1386	1566	1642
LOC115537228	2106	2402	2555	2008	1845	2826	2127	2235
LOC115537019	659	544	911	564	505	611	696	444
LOC115538651	727	674	630	564	479	599	554	523
LOC115538267	57	192	81	153	104	137	110	104
kbtbd12	1003	743	646	766	715	875	301	216

Gene counts

How do we get gene expression data, and what does it look like?

Samples

Treatment *Control*

functional info for many gene

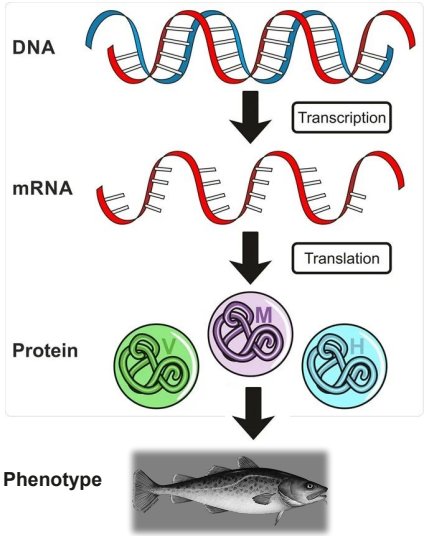
Genes

Gene counts

gene_gadmor	PCG001	PCG004	PCG011	PCG015	PCG017	PCG020	PCG029	PCG035	spid	species	evalue	protein_names
ND1	8276	8202	8327	7750	5508	4351	5673	6385	P55779	GADMO	0.00e+00	NADH-ubiquinone oxidoreductase chain 1 (EC 7.1.1.2) (NA
ND2	39142	43297	57032	31681	33489	30275	38921	36460	P55780	GADMO	2.40e-158	NADH-ubiquinone oxidoreductase chain 2 (EC 7.1.1.2) (NA
COX1	555463	631876	917827	551062	378628	403956	438595	455514	Q36775	GADMO	0.00e+00	Cytochrome c oxidase subunit 1 (EC 7.1.1.9) (Cytochrome c
COX2	315625	309958	493176	294979	189346	229132	227614	219894	Q37741	GADMO	1.12e-132	Cytochrome c oxidase subunit 2 (EC 7.1.1.9) (Cytochrome c
ATP6	82892	105415	89100	73253	77770	58061	86703	78859	P55778	GADMO	2.60e-104	ATP synthase subunit a (F-ATPase protein 6)
COX3	165275	189321	235193	135559	112870	81377	135462	126615	P55777	GADMO	5.84e-152	Cytochrome c oxidase subunit 3 (EC 7.1.1.9) (Cytochrome c
ND3	10710	13595	10404	7013	10244	2567	9792	8231	P15957	GADMO	1.48e-32	NADH-ubiquinone oxidoreductase chain 3 (EC 7.1.1.2) (NA
ND4L	19364	31196	42855	16261	20566	17877	21021	16269	P23633	GADMO	1.08e-41	NADH-ubiquinone oxidoreductase chain 4L (EC 7.1.1.2) (N
ND4	37081	42648	67118	30275	29274	37939	35681	34155	P55781	GADMO	0.00e+00	NADH-ubiquinone oxidoreductase chain 4 (EC 7.1.1.2) (NA
ND5	16299	21803	2823			19960	19262	18187	P55782	GADMO	0.00e+00	NADH-ubiquinone oxidoreductase chain 5 (EC 7.1.1.2) (NA
ND6	4102	3787	318			1774	3772	3391	P55783	GADMO	1.86e-42	NADH-ubiquinone oxidoreductase chain 6 (EC 7.1.1.2) (NA
CYTB	176843	211635	27739			114784	137070	144369	Q37080	GADMO	0.00e+00	Cytochrome b (Complex III subunit 3) (Complex III subunit
LOC115539476	260	228	333	219	270	266	312	266	Q99MK9	MOUSE	1.66e-27	Ras association domain-containing protein 1 (Protein 123F
LOC115539709	890	849	919	777	561	461	945	1000	Q9WVF8	MOUSE	5.41e-27	Tumor suppressor candidate 2 (Fusion 1 protein) (Fus-1 pr
LOC115538781	586	596	576	630	417	450	704	730	Q12891	HUMAN	3.51e-104	Hyaluronidase-2 (Hyal-2) (EC 3.2.1.35) (Hyaluronoglucosam
abhd14a	1197	1381	1629	1031	774	757	1158	1108	Q1LV46	DANRE	4.34e-34	Protein ABHD14A (EC 3.-.-.-) (Alpha/beta hydrolase domai
acy1	1626	1244	1670	1245	1224	1386	1566	1642	Q6PTT0	RAT	2.57e-16	Aminoacylase-1B (ACY-1B) (EC 3.5.1.14) (ACY IB) (N-acyl-L-
LOC115537228	2106	2402	2555	2008	1845	2826	2127	2235	Q6PHS9	MOUSE	1.07e-34	Voltage-dependent calcium channel subunit alpha-2/delta
LOC115537019	659	544	911	564	505	611	696	444	Q90339	CYPCA	1.94e-172	Myosin heavy chain, fast skeletal muscle
LOC115538651	727	674	630	564	479	599	554	523	Q9NXG6	HUMAN	5.61e-20	Transmembrane prolyl 4-hydroxylase (P4H-TM) (EC 1.14.11
LOC115538267	57	192	81	153	104	137	110	104	Q8CIW6	MOUSE	3.91e-23	Solute carrier family 26 member 6 (Anion exchange transp
kbtbd12	1003	743	646	766	715	875	301	216	Q3ZB90	DANRE	0.00e+00	Kelch repeat and BTB domain-containing protein 12 (Kelch

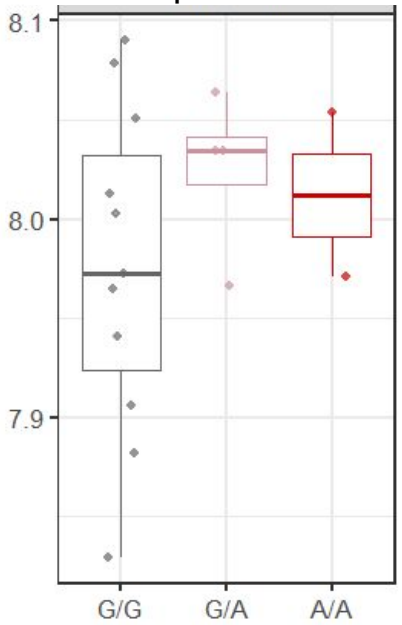


Genetic variation in prkci associated with HSI, higher prkci expression, higher liver lipid & growth rates

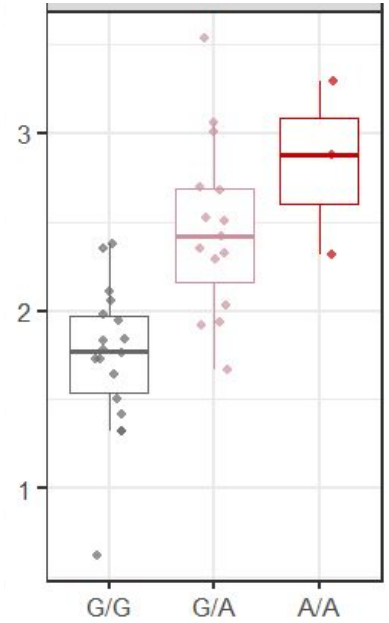


Adapted from: udaix/Shutterstock.com

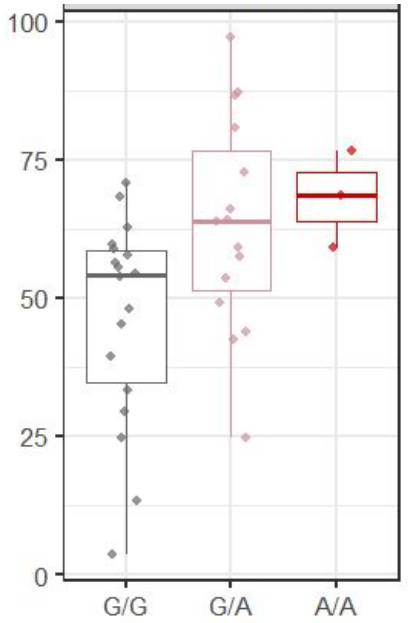
prkci gene expression



HSI

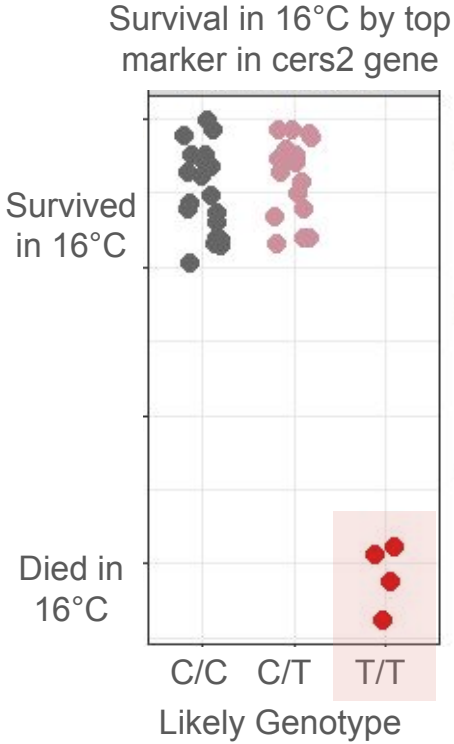


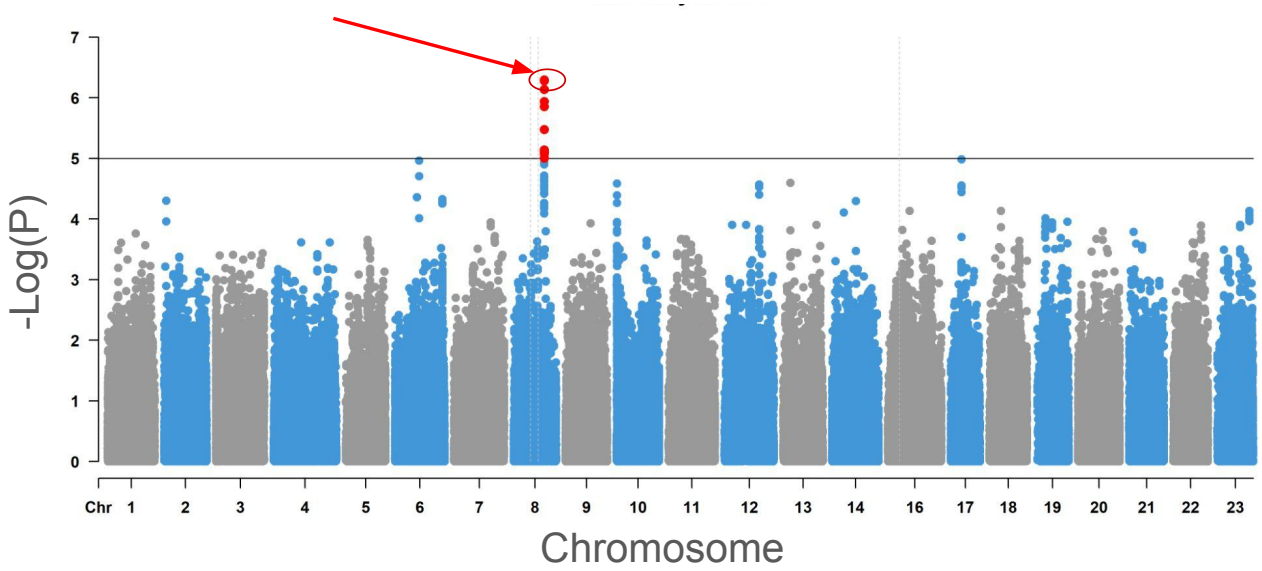
Growth rate (ww)





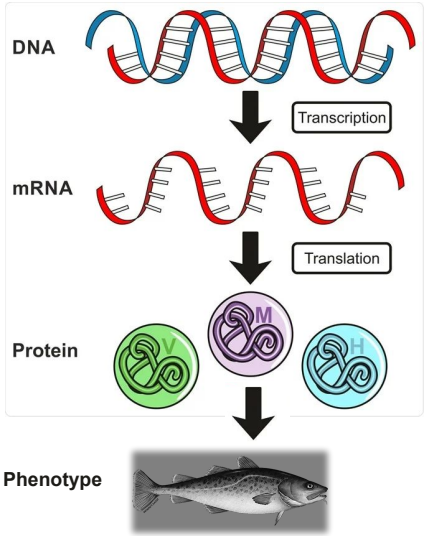
SNP markers in 1 gene, ceramide synthase 2 (cers2) associated with **mortality in warming (16°C)** in Pacific cod juveniles





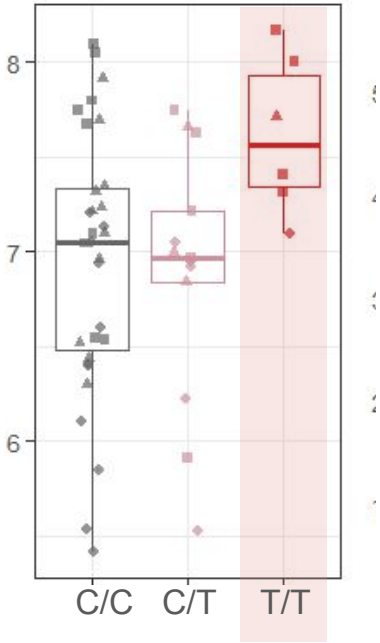


Genetic variation in cers2 associated with mortality in warming, higher cers2 expression, lower liver lipid & growth rates



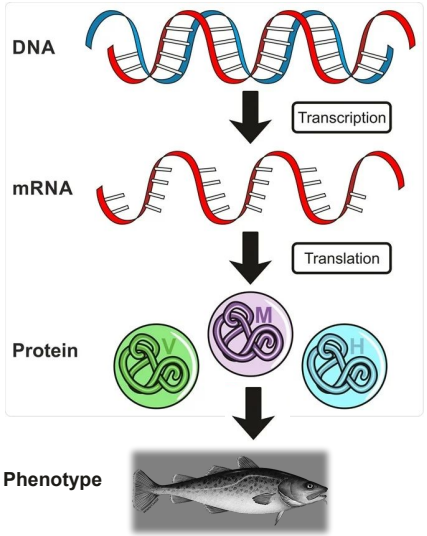
Adapted from: udaix/Shutterstock.com

Ceramide synthase 2 expression in liver



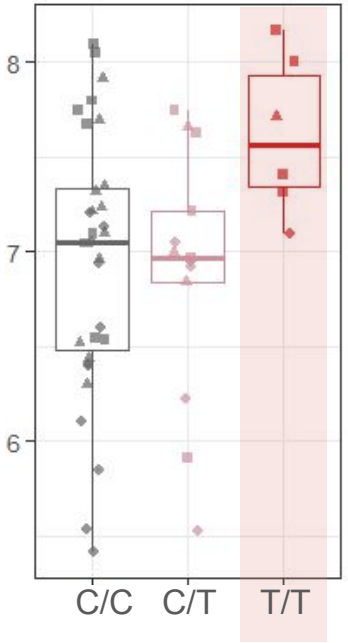


Genetic variation in cers2 associated with mortality in warming, higher cers2 expression, lower liver lipid & growth rates

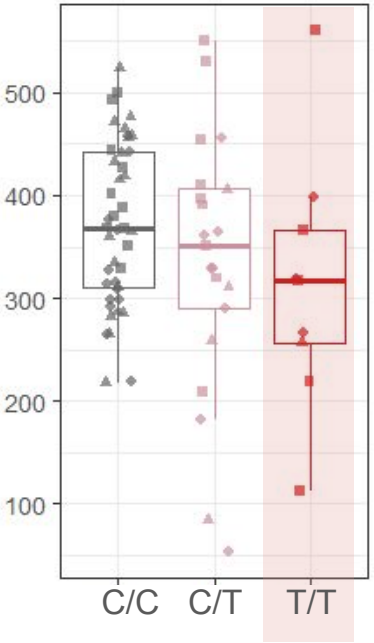


Adapted from: udaix/Shutterstock.com

Ceramide synthase 2 expression in liver

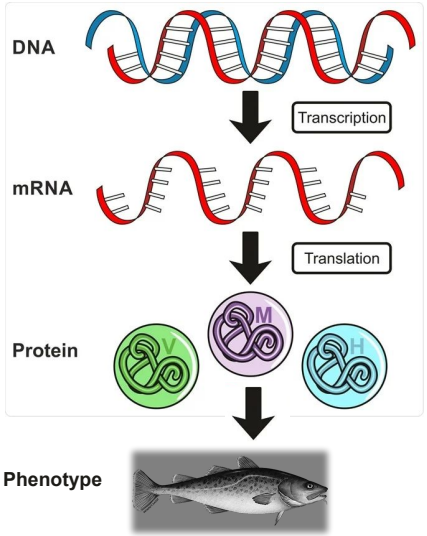


Lipid concentration in liver



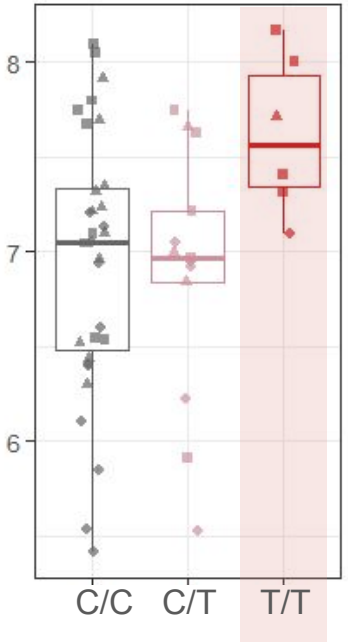


Genetic variation in cers2 associated with mortality in warming, higher cers2 expression, lower liver lipid & growth rates

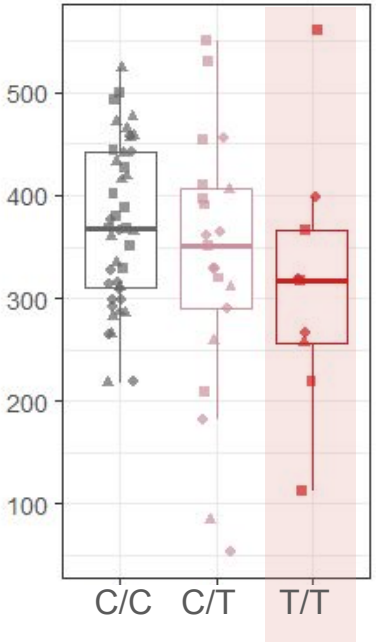


Adapted from: udaix/Shutterstock.com

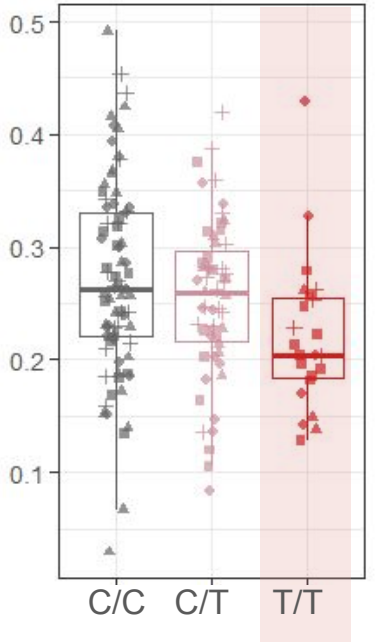
Ceramide synthase 2 expression in liver



Lipid concentration in liver



Growth rate during acclimation (all at 6°C)





Genetic variation in cers2 associated with mortality in warming, higher cers2 expression, lower liver lipid & growth rates

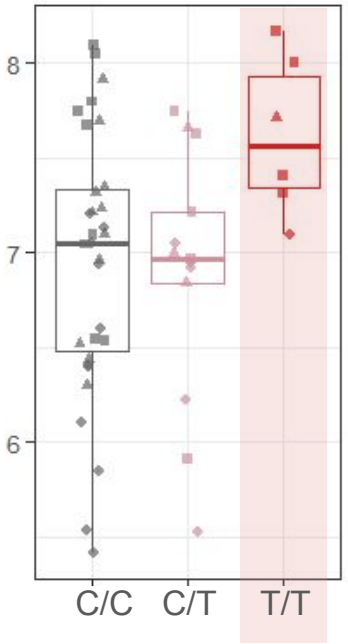
Cers2 synthesizes long-chain fatty acids (ceramides), involved in membrane structure, stress signaling

Who can survive heat waves?

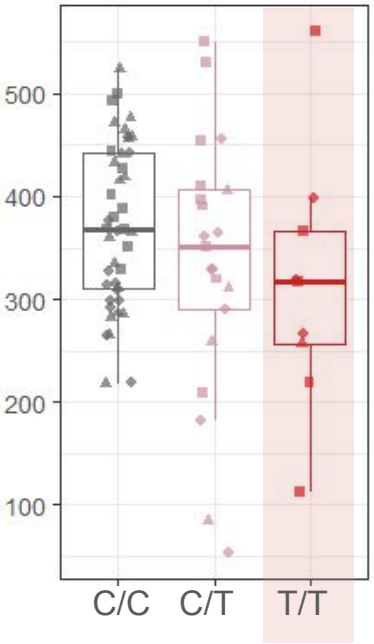
Maybe not young cod with T/T in cers2 marker

… *not necessarily causal, and traits are polygenic!!*

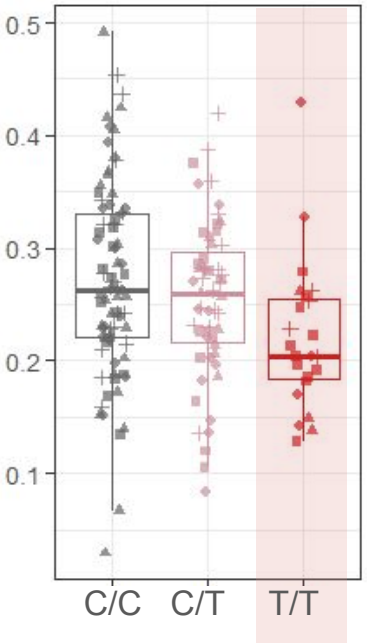
Ceramide synthase 2 expression in liver



Lipid concentration in liver

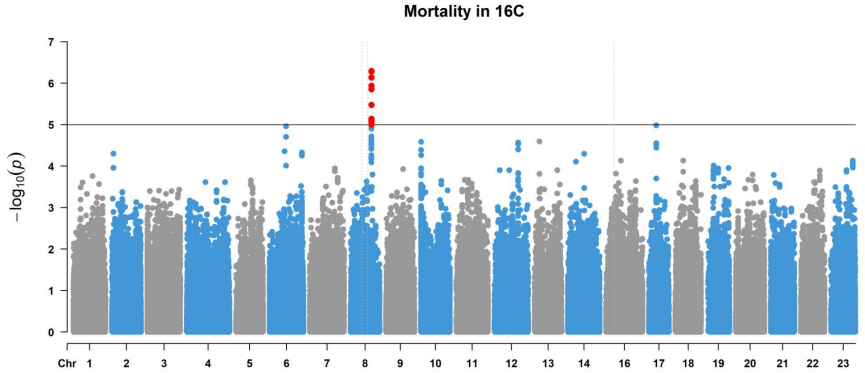


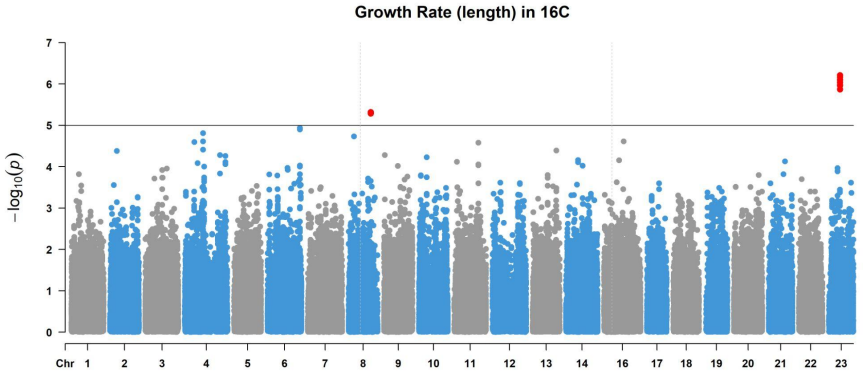
Growth rate during acclimation (all at 6°C)

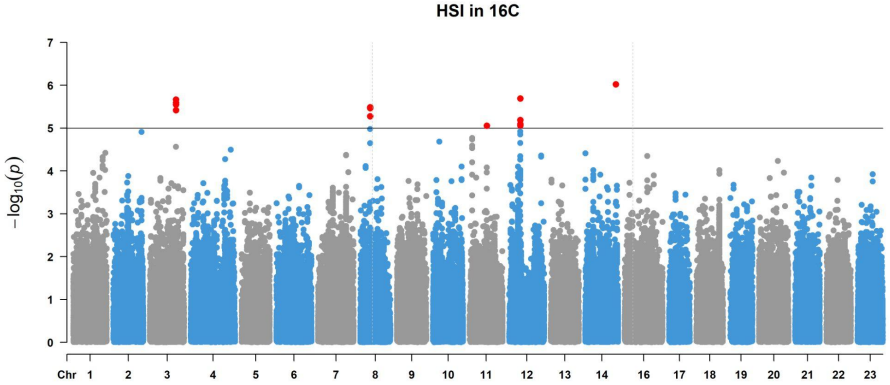


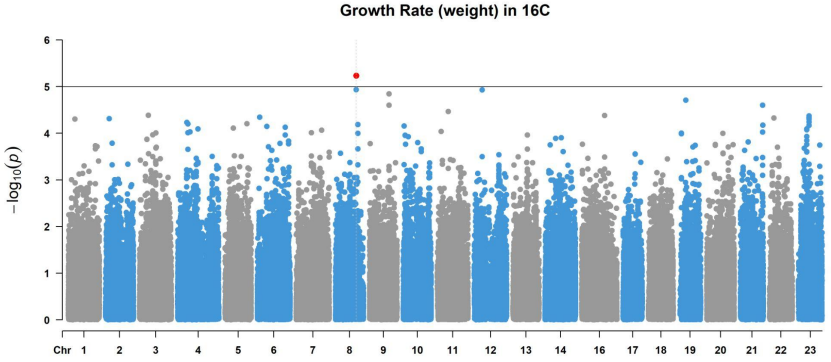


Next step: Calculate polygenic score for beneficial traits



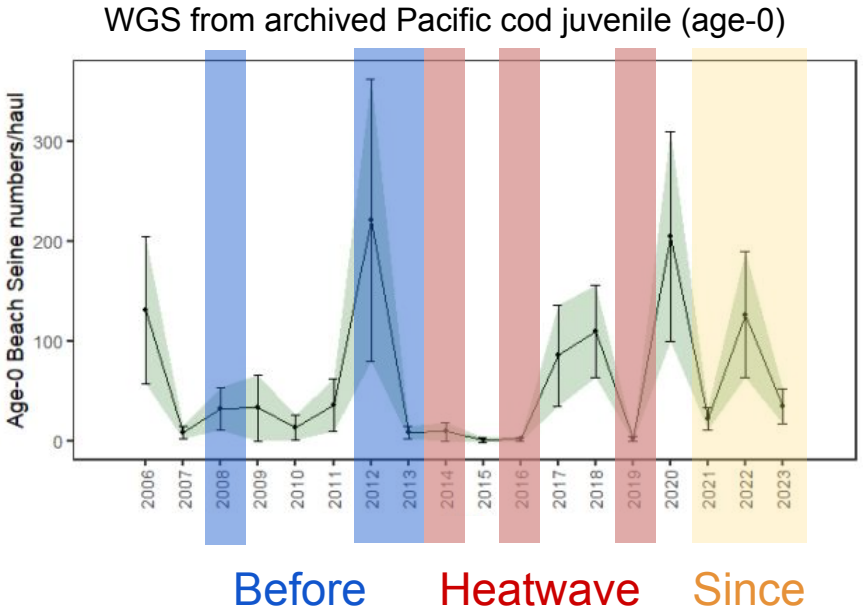








Who actually survived heat waves? TBD!



Genetic Time Series

- Juveniles from nine distinct years, spanning heatwaves (2008-2023)
- ~40 per year, all in July off Kodiak
- Analysis: track allele frequencies, polygenic scores over heatwaves
- **Signals of selection for trait-associated markers?**



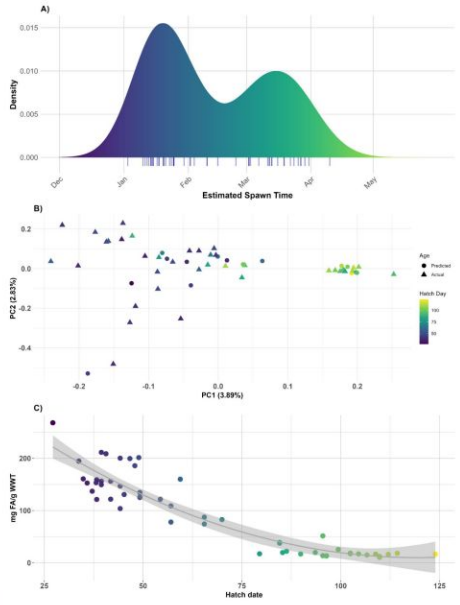
Pacific cod juvenile genome-to-phenome temperature study

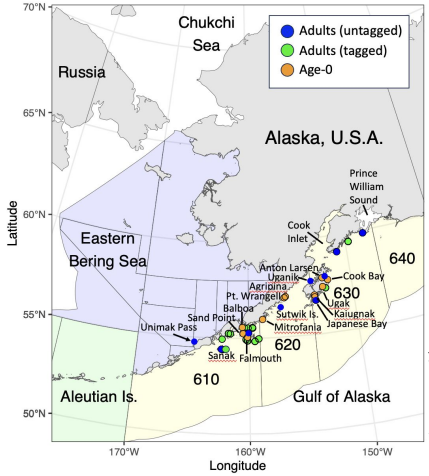
Experimental results:

- Juvenile cod growth & energy allocation is highly sensitive to temperature.
  - *Interestingly, higher T_{max} for growth compared to 10 years prior*
- **Supports hypothesis that low recruitment in GOA during heat waves likely related to high mortality rates through juvenile stage,** and **low overwintering survival due to fewer lipid reserves.**
- Stock assessments (AI, GOA) recently **adjusted natural mortality rates** during heatwave years to account for temperature

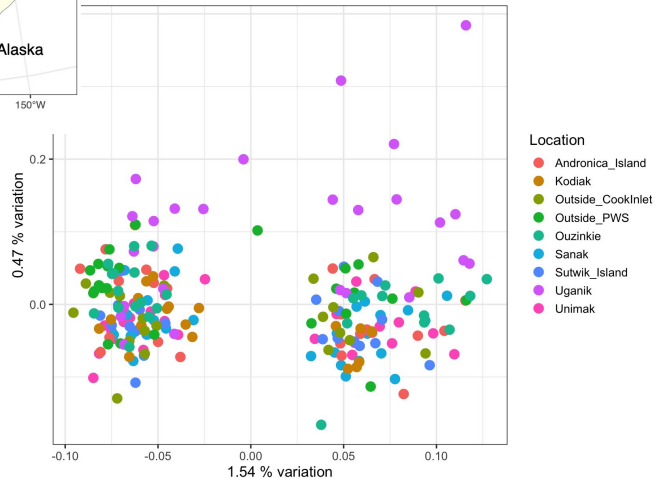


There is variation within wGOA/EBS Pacific cod, which is good!





New research! *There are two genetically distinct groups in GOA with varying depth (and possibly temperature) preferences - "ecotypes"*



First evidence of genetically distinct winter- and spring-spawning components of Pacific cod in the Gulf of Alaska

Authors: Benjamin Laurel, Jessica Miller, Louise Copeman, Alisa Abookire, Laura Spencer, Sara Schaal, Mary Beth Rew Hicks, Nicholas Strait, Lauren Rogers, and Ingrid Spies | AUTHORS INFO & AFFILIATIONS

Publication: Canadian Journal of Fisheries and Aquatic Sciences • 16 January 2026 • https://doi-org.offcampus.lib.washington.edu/10.1139/cjfas-2025-0168

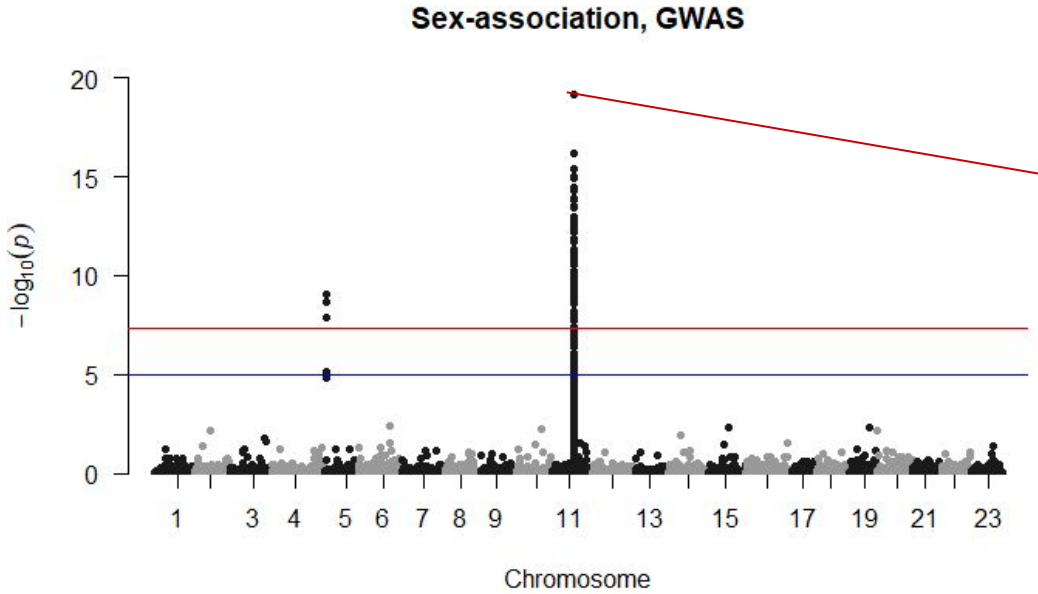


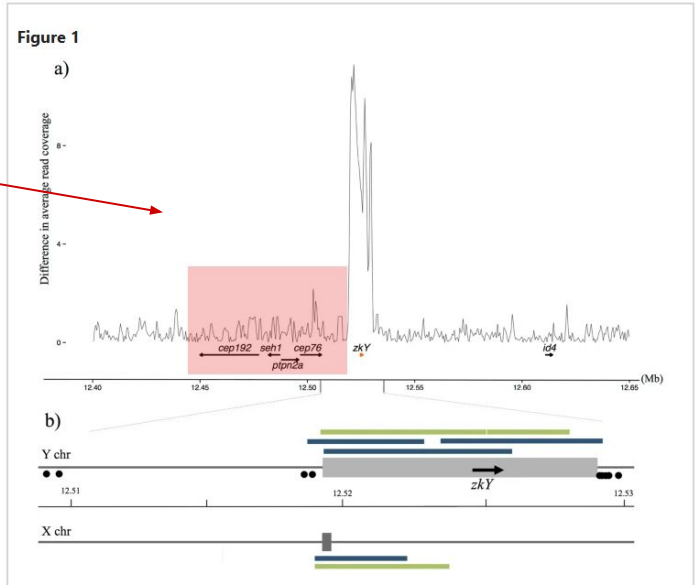
Growing Pacific cod ‘Omics resources

- Chromosome-level genome
- Genetic assignment of population, within-population groups (“ecotypes”)
- SNP Reference Panel for Pacific cod in GOA / EBS - *for imputation, etc.*
- Liver gene expression + liver lipid profiles
- Genetic assignment of sex
- Hoping to fund epigenetic aging project

Genetic sex determination in Pacific cod, follows X/Y system

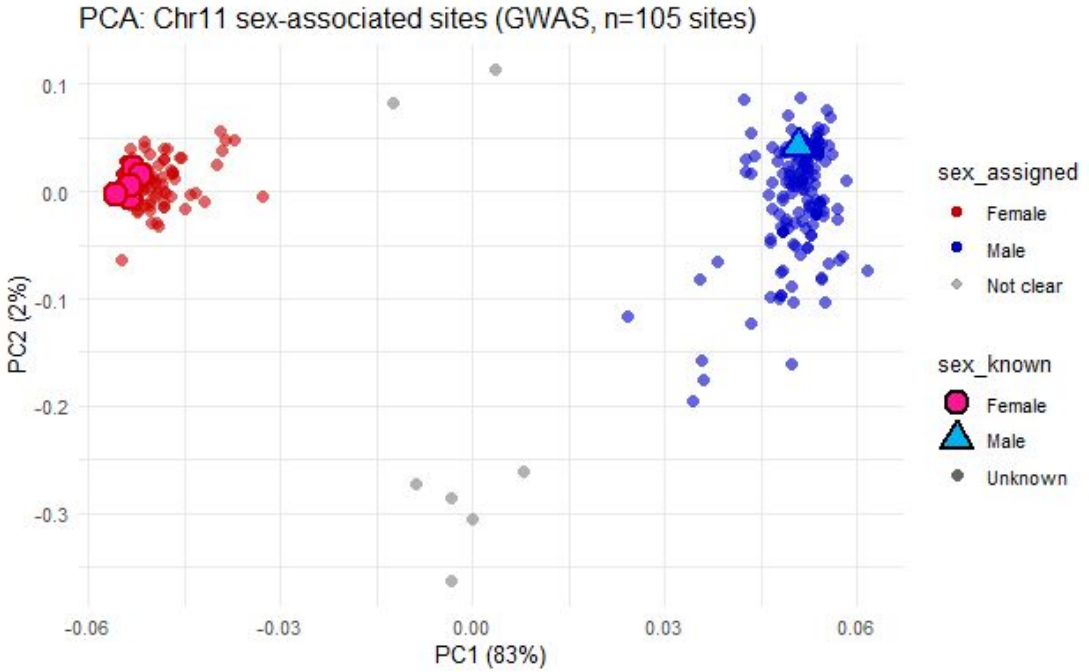
Sex markers in same region of Chromosome 11 as in Atlantic cod

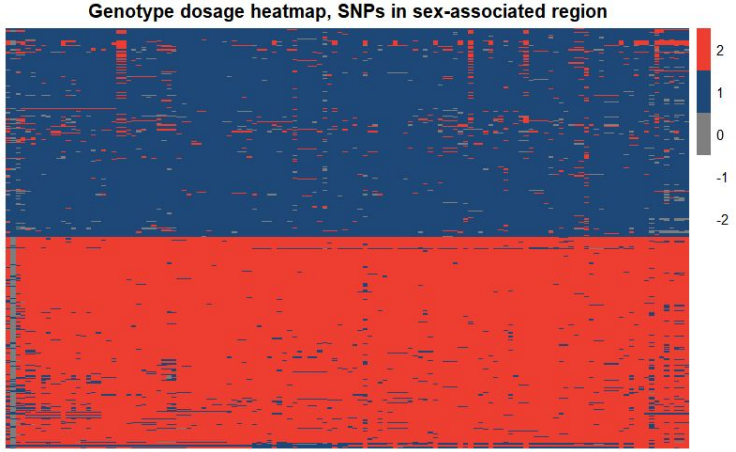




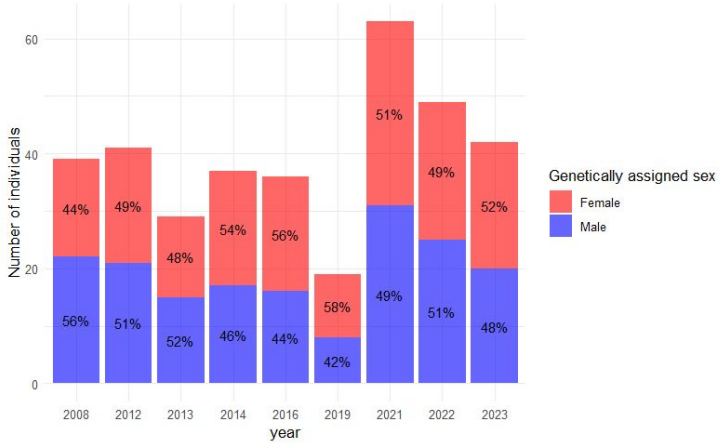
Pacific cod sex can be determined genetically, we are doing this now with lcWGS

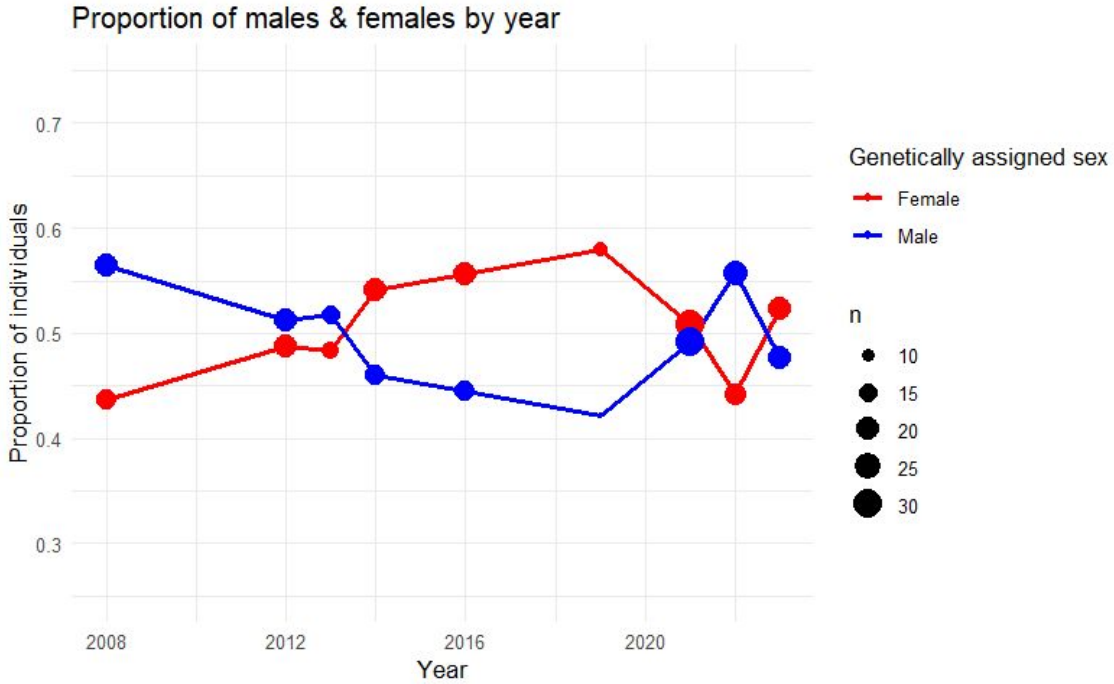
Applications: Tagged cod, early life stages (larvae, age-0 juveniles), experiments





Pacific cod juvenile sex ratios - could they be female skewed during heatwaves?





Contributors & Funders

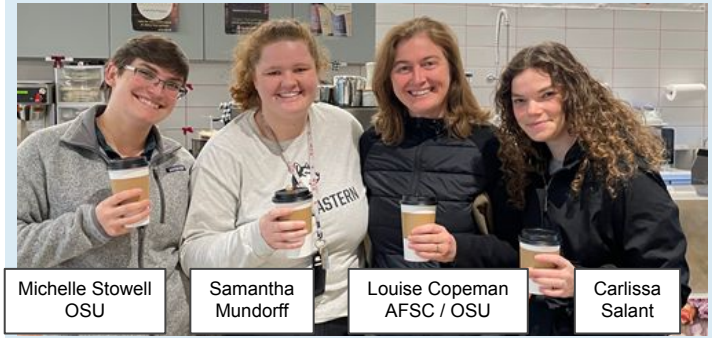
Juvenile Collection & Experimental Data


Ben Laurel, AFSC


Mary Beth Rew Hicks, AFSC


Paul Iseri, AFSC

Lipid Data


Michelle Stowell OSU

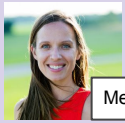
Samantha Mundorff

Louise Copeman AFSC / OSU

Carlissa Salant



‘Omics Data & Bioinformatics


Me


Ingrid Spies, AFSC


Sara Schaal, AFSC & UO


Wes Larson, AFSC


Steven Roberts, UW


Laura Timm, AFSC


Kathleen Durkin, UW